IGIS
INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS

GSP
GEOLOGICAL SURVEY OF PAKISTAN



INTERNATIONAL TRAINING WORKSHOP

Remote Sensing *for Disaster Management*

From satellite constellations to UAV systems: an integrated observation chain for emergency response and loss assessment

18–20 November 2026
IGIS, SCEE, NUST · Islamabad

Programme booklet

Two scales. One observation chain.

During a monsoon flood, responders need two answers at once: *how far has the water spread, and who is affected?* Then: *which embankments breached, which bridges failed, how many homes are lost?*

> No single sensing platform can address both. Used in a deliberate sequence, satellites and UAVs create an observation chain far more effective than either alone.

Over two and a half days you will work through that chain, using openly licensed satellite data, browser-based cloud computing and standard commercial UAVs, all reproducible with equipment ordinarily available to a Pakistani institution.

BY THE CLOSE OF DAY 3 YOU WILL HAVE PRODUCED

1. **A flood extent map** from Sentinel-1 SAR
2. **An exposure table** of population, cropland and roads
3. **A UAV orthomosaic and DSM** from imagery you helped fly
4. **A damage assessment** quantified from that model
5. **A rapid situation brief** for an emergency operations centre

PROGRAMME AT A GLANCE

From the wide view to the close-up.

01

WEDNESDAY 18 NOVEMBER · 08:30–17:00

Foundations, wide-area assessment & mission design

Modules 1–3 · Lab 0 · Lab 1A/1B flood mapping and exposure · Lab 2 flight planning

02

THURSDAY 19 NOVEMBER · 08:30–17:00

Field operations & UAV data processing

Field flights at the NUST field site · Lab 3A photogrammetry · Lab 3B 3D modelling and damage quantification

03

FRIDAY 20 NOVEMBER · 09:00–12:15

Integration, case studies & rapid-response simulation

Module 4 · three Pakistani case studies · capstone simulation · closing and certificates

2½	15	30	4
days	contact hours	participants	expert lecturers

DAY 1 · WEDNESDAY 18 NOVEMBER

Foundations & wide-area assessment

MORNING

08:30	Registration and tea	Badge collection, laptop check and Earth Engine login verification.
09:00	**Opening ceremony**	Welcome address, inauguration and course objectives.
09:20	**Module 1 · Foundations of remote sensing across scales**	Passive and active sensing; the resolution quartet; swath versus resolution; what “large-scale” changes; the observation continuum from satellites to ground surveys; Earth observation across the disaster management cycle.
10:10	**Module 2 · Data access and international emergency mapping**	Sentinel, Landsat, MODIS, VIIRS, GPM IMERG and global DEMs; Analysis-Ready Data; Copernicus Data Space and the Earth Engine catalogue; the International Charter, Copernicus EMS, Sentinel Asia and UN-SPIDER, and how a Pakistani agency requests activation.
10:55	Tea break	
11:10	**Module 3 · UAV systems, regulation and payloads**	UAV system composition; fixed-wing, multirotor and VTOL platforms; optical, multispectral, thermal and LiDAR payloads; regulations, authorities and operating procedures in Pakistan.
12:00	Lunch and prayers	

DAY 1 · WEDNESDAY 18 NOVEMBER

Mapping the flood & planning the flight

AFTERNOON

13:30	**Lab 0 · Environment check** Earth Engine Code Editor orientation, shared scripts, a first run and login troubleshooting. Offline data package for anyone without access.
14:00	**Lab 1A · Rapid flood mapping from Sentinel-1 SAR** Pre- and post-event VV imagery, speckle filtering, change-detection ratio and thresholding, terrain masking with slope and HAND. Output: flood extent for the 2022 Sindh floods.
15:30	Tea break
15:45	**Lab 1B · Exposure and impact quantification** Flood extent against WorldPop, ESA WorldCover and OpenStreetMap to build a district-wise exposure table; error sources and where they justify an aerial survey.
16:20	**Lab 2 · Flight planning and mission design** Area, sensor, overlap, ground sample distance and safety; avoiding restricted airspace; altitude, speed and route generation. Each group plans the mission it flies on Day 2.
17:00	Close of Day 1 Field safety and pre-flight briefing issued.

DAY 2 · THURSDAY 19 NOVEMBER

Field operations & UAV data processing

08:30	**Transfer and pre-flight briefing**	To the NUST field site. Airspace, weather and safety limits, crew roles and go/no-go; aircraft checks and ground control layout.
09:00	**Field practical · UAV flight operations**	Supervised stations: manual handling, emergency take-off and landing, return-to-home; the planned mission; orthophoto, oblique and thermal acquisition; GCP survey.
11:30	**Data offload and quality check**	Coverage, overlap and image quality; flight log; transfer to workstations.
12:00	Lunch and prayers	Return to IGIS.
13:30	**Lab 3A · Photogrammetric processing**	POS data, pre-processing, alignment, dense reconstruction, DSM and orthomosaic, georeferencing and accuracy.
15:15	Tea break	
15:30	**Lab 3B · 3D modelling and damage quantification**	Oblique modelling; change detection between epochs; landslide displacement; building damage and road length, reconciled with the Lab 1 exposure table.

If weather or airspace prevents flying, a pre-flown reference dataset keeps the labs on schedule.

DAY 3 · FRIDAY 20 NOVEMBER · HALF DAY

Integration, case studies & rapid response

09:00 **Module 4 · Integrating the observation chain**
Nested assessment from satellite screening to UAV tasking and ground validation; handover points, authority and latency; uncertainty and defensible figures.

09:35 **Applied case studies**
Attabad Lake flood risk · Shisper Lake GLOF with HEC-RAS 2D · Jhang riverine flooding, ENSO and the monsoon.

10:15 Tea break
Scenario cards and data packages distributed.

10:30 **Capstone · Rapid-response simulation**
Six groups, 45 minutes, a four-slide brief: event footprint, exposure table, UAV tasking plan and confidence statement. Scenarios include Dera Ghazi Khan floods, a Hunza GLOF, a Muzaffarabad earthquake, Tharparkar drought, Karachi urban flooding and Sindh reconstruction.

11:45 **Communication, summary and closing**
Pak-NSDI and data sharing; self-study packages and online clinic; remarks by dignitaries, certificates and group photograph.

12:15 Close
Ahead of Friday congregational prayers.

LEARNING OUTCOMES

What you'll be able to do.

01 Choose platform, sensor and resolution for a hazard and reporting scale.

02 Access free satellite archives and request international emergency mapping.

03 Map flood extent from Sentinel-1 SAR and quantify exposure.

04 Select a UAV platform and payload against mission requirements.

05 Follow Pakistan's UAV regulations, permissions and flight-safety procedures.

06 Plan and fly a survey mission for orthophoto, oblique and thermal data.

07 Process imagery to an orthomosaic, a DSM and an oblique 3D model.

08 Quantify damage and reconcile it with satellite exposure estimates.

09 Design a nested multi-scale workflow with stated uncertainty.

10 Deliver a rapid situation brief under time pressure.

EXPERT LECTURERS

Your lecturers.

NY

Dr. NIE Yong

Professor · Institute of Mountain Hazards and Environment (IMHE), CAS

Director, Mountain Science Data Center. Glacier and glacial lake hazards and remote sensing; 50+ papers including ESI highly cited work; risk assessment for the China–Pakistan Economic Corridor.

WW

Engr. WANG Wen

Assistant Engineer · Mountain Science Data Center, IMHE, CAS

Remote sensing for mountain hazards and the cryosphere; extensive UAV fieldwork and processing, from flight planning to orthomosaics, DEMs and 3D models.



Dr. Javed Iqbal

Professor · SCEE – IGIS, NUST

Three decades in GIS, remote sensing and hazard mapping of landslides, glacier avalanches and flash floods; 200+ scholarly contributions.

SK

Dr. Shahid Nawaz Khan

Assistant Professor · SCEE – IGIS, NUST

Explainable machine learning for Earth observation, SAR–optical fusion, Earth Engine and UAV remote sensing; FAA Part 107 remote pilot.

BEFORE YOU ARRIVE

Come ready to work.

The schedule is compressed, so there is no slack to recover from a failed login on the day.

- [ ] **Google Earth Engine access** registered and verified with a linked Cloud project. Allow 7–10 days.
- [ ] **Laptop with Google Chrome.**
- [ ] **QGIS 3.34 LTR** installed, with 5 GB free disk space.
- [ ] **UAV ground-station app** on a charged phone or tablet, with a cable.
- [ ] **Field clothing for Day 2:** closed-toe shoes, sun protection and a hat.
- [ ] **Pre-reading pack**, circulated two weeks ahead.
- [ ] **Signed field-safety acknowledgement**, handed in at registration.

VENUE & CONTACT

Where to find us.

LECTURES & LABS · ALL THREE DAYS

Institute of Geographical Information Systems

SCEE, NUST Sector H-12, Islamabad 44000

UAV FLIGHTS · DAY 2 MORNING

NUST Field Site

Near the Saddle Club, NUST H-12 · transfer from IGIS at 08:30

NUST_CPJRC_WRKSHP@igis.nust.edu.pk
+92 322 529 2331 · +92 379 375 724

NOTES

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INSTITUTE OF GEOGRAPHICAL INFORMATION SYSTEMS

GSP GEOLOGICAL SURVEY OF PAKISTAN

China–Pakistan Joint Research Center on Earth Sciences
Institute of Geographical Information Systems, School of Civil and Environmental Engineering, NUST
Geological Survey of Pakistan

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